

POLITECNICO DI MILANO

TOPOLOGY OPTIMIZATION

Topology Optimization of a 2D Truss Structure

Assignment 1

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1 Introduction

Structural optimization, one of the fundamental disciplines of engineering design, focuses on designing material assemblies in the most efficient manner under specified loads and boundary conditions. Various structural performance indicators, such as stiffness, weight, stress, and natural frequencies, are used to quantitatively define what constitutes a "most efficient" design. This project focuses on sizing optimization, one of the most fundamental branches of structural optimization. In sizing optimization, the physical dimensions of the elements, such as cross-sectional area or thickness, are optimized as design variables, while the topology and overall shape of the structure are kept constant. This study addresses the size optimization of a 2D truss structure. The truss structure resembles a cantilever beam consisting of 77 nodes and 200 bar elements. The design variables ($\mathbf{x}$) are defined as the cross-sectional areas (A_j) of each bar element. The main objective of the problem is to minimize structural flexibility (structural compliance, C), which is equivalent to maximizing the overall stiffness of the structure. Elasticity is defined as the work done by external forces and is expressed by the following formula:

$$\mathbf{C} = \mathbf{F}^T \mathbf{u} \quad (1)$$

Here, $\mathbf{F}$ represents the global vector of external loads, and $\mathbf{u}$ represents the vector of nodal displacement under these loads. Optimization is performed under two fundamental constraints. The first is the global volume constraint, which imposes an upper bound on the total volume of the structure (V). This constraint is formulated such that the target volume does not exceed V_0 :

$$V - V_0 \leq 0 \quad (2)$$

The target volume V_0 is defined as a certain ratio (α , volume fraction) of the maximum volume the structure can occupy ($V_{\max}$).

$$V_0 = \alpha V_{\max} \quad (3)$$

Second, for each design variable x_j (the cross-sectional area of the j -th element), box constraints are defined, resulting from manufacturability and physical requirements:

$$l_b \leq x_j \leq u_b \quad (4)$$

For this project, the lower and upper bounds are set as $l_b = 0.0001 \text{ mm}^2$ and $u_b = 800 \text{ mm}^2$, respectively. To solve such a complex and generally non-convex optimization problem, a gradient based iterative algorithm is adopted. The chosen solution approach is the CONLIN (Convex Linearization) method. This method transforms the original problem into explicit, convex and separable subproblems that are easier to solve at each iteration step. These subproblems are efficiently solved using Lagrange dual formulation. In this report, the performance of the methodology and the effect of different material limitations on the design will be evaluated for four different volume fraction scenarios $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$.

2 Problem Definition

This section details an optimization problem for a two-dimensional (2D) truss structure as a concrete application of structural optimization principles. The primary objective is to maximize the structural stiffness within defined material and geometric constraints, thereby achieving the most efficient material distribution.

2.1 Boundary Conditions and External Loads

The structure's behavior is determined by the boundary conditions and external loads acting on it:

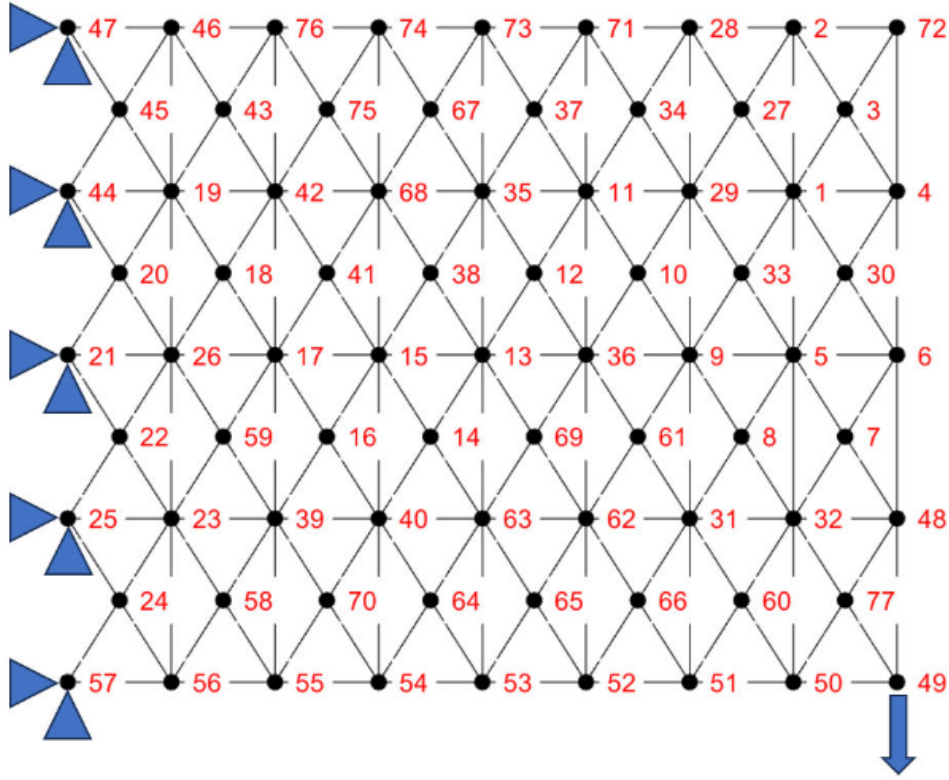


Figure 1: 2D truss model: fixed nodes on the left; vertical load at node 49 (bottom-right).

- **Fixed Boundary Conditions:** Nodes 47, 44, 21, 25, and 57, located along the left vertical edge of the truss, are completely fixed to the ground by hinged supports. This means that both horizontal and vertical displacement degrees of freedom of these nodes are blocked ($u_x = u_y = 0$).
- **Applied External Load:** A concentrated vertical downward load of 6000 N is applied to node 49, located at the bottom right corner of the structure [?]. This load is considered constant throughout the optimization process.

2.2 Optimization Problem Formulation

The optimization problem is mathematically formulated as the minimization of an objective function under certain constraints as follows:

- **Objective Function:** The main objective of the problem is to minimize the structure's compliance, i.e., the work done by external forces. Minimizing compliance is equivalent to maximizing structural stiffness. The objective function $C(\mathbf{x})$ is expressed as follows:

$$\min_{\mathbf{x}} C(\mathbf{x}) = \mathbf{F}^T \mathbf{u}(\mathbf{x}) \quad (5)$$

Here $\mathbf{F}$ represents the global external load vector, and $\mathbf{u}(\mathbf{x})$ represents the global nodal displacement vector calculated depending on the design variables ($\mathbf{x}$).

- **Design Variables:** The basis of the optimization is the cross-sectional area x_j of each element j . These variables are represented as the vector $\mathbf{x} = [x_1, x_2, \dots, x_{200}]^T$.
- **Constraints:**

- *Global Volume Constraint:* The total material volume of the structure, $V(\mathbf{x})$, is constrained by a predetermined target volume V_0 :

$$V(\mathbf{x}) = \sum_{j=1}^{200} L_j x_j \leq V_0 \quad (6)$$

Where L_j is the length of the j -th element. The target volume V_0 is defined as a certain ratio (α) of the maximum theoretical volume ($V_{\max}$) when the cross-sectional area of all elements is at the upper boundary (u_b): $V_0 = \alpha V_{\max}$. In this project, the values $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$ are investigated for four different material scenarios.

- *Box Constraints:* Lower and upper bounds are available for each design variable x_j :

$$l_b \leq x_j \leq u_b \quad (7)$$

In this study, the lower bound is set to $l_b = 0.0001 \text{ mm}^2$ and the upper bound is set to $u_b = 800 \text{ mm}^2$ [?]. Using a small lower bound close to zero is standard practice to prevent the stiffness matrix from becoming singular during analysis.

2.3 Optimization Approach

To solve this non-linear and non-convex optimization problem, a gradient-based algorithm is used, where the problem is transformed into an approximate subproblem at each iteration. As stated in the project description, this approximation is performed with the CONLIN (Convex Linearization) method. CONLIN creates a high-quality, convex subproblem by linearizing the objective and constraint functions with respect to the design variables themselves (direct variables) or their inverse (inverse variables, $1/x_j$). This strategy ensures stable and rapid convergence, particularly in stiffness optimization.

This convex and separable subproblem is solved efficiently using Lagrangian duality at each iteration step. The ‘dphi’ function in the MATLAB code applies the Karush-Kuhn-Tucker (KKT) conditions, which are derived from the solution of this dual problem and calculate the optimal new design variables (‘xnew’). For this process to work, the ****sensitivities****, which are the derivatives of the objective function and constraints with respect to the design variables, must be calculated at each step.

3 Computational Workflow

This section details the practical implementation steps and underlying processes used to solve the minimum flexibility (maximum stiffness) problem defined for a cantilever-type truss structure. The workflow is based on an iterative optimization loop developed in the MATLAB programming environment and based on the sequential approximation technique.

3.1 MATLAB Implementation Overview

MATLAB is a widely preferred platform for solving structural optimization problems, particularly due to its strength and flexibility in matrix-based operations. The approach used in this study adopts a modular code structure. The main file (main5m.m) manages a series of specialized functions that perform tasks such as Finite Element Analysis (FEA), sensitivity calculation, and design update. The code achieves memory efficiency by using sparse matrices (`sparse`), efficiently solves linear systems ($Ku=F$) with the `mldivide` (`\`) operator, and handles the non-linear optimization subproblem via the `fsolve` function. This approach allows transparently applying the underlying physics and mathematical steps of the problem.

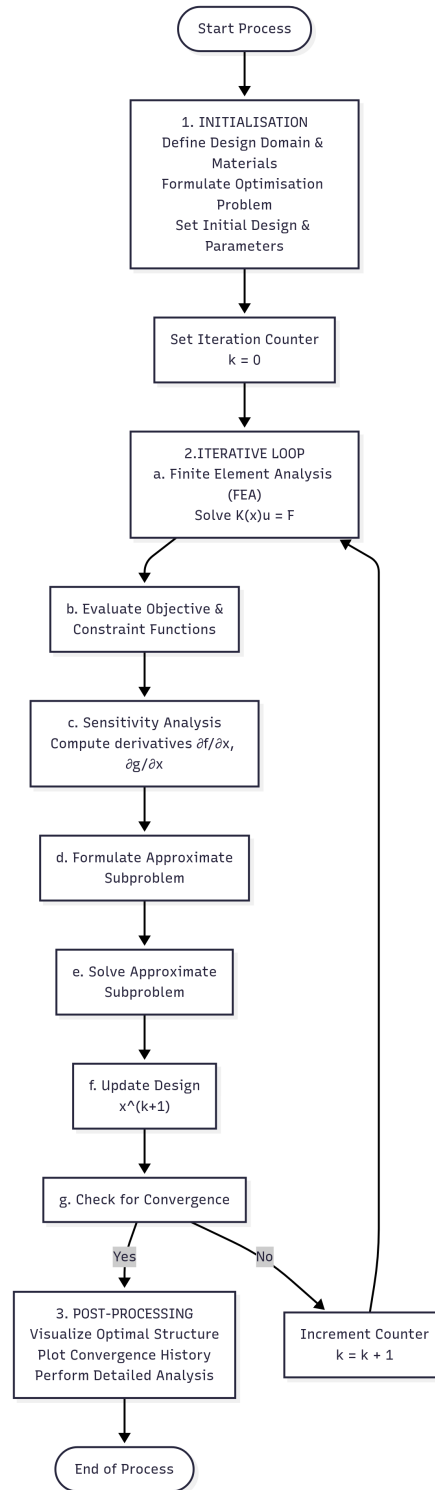


Figure 2: General workflow diagram of the structural optimization problem

3.2 Algorithm Initialization and Main Loop

Before the optimization cycle begins, the parameters of the problem and the algorithm are defined:

- **Data Loading and Structure Definition:** Files containing the structure's nodal coordinates (`nodes.txt`) and element connections (`elements.txt`) are loaded into the MATLAB workspace. Young's Modulus (E) is defined as the material property.
- **Initialization of Design Variables:** The design variables, cross-sectional areas (x), are initially

assigned an equal value for all elements. This initial vector, called ‘xold’, is calculated to exactly satisfy the total volume constraint. This ensures that the optimization begins with a feasible solution. Additionally, lower ($l_b = 0.0001 \text{ mm}^2$) and upper ($u_b = 800 \text{ mm}^2$) cross-sectional area limits are determined for each element. A lower limit greater than zero prevents singularities that may occur in the stiffness matrix.

- **Definition of Constraints:** Volume ratios (α) are defined for the four scenarios to be examined. The total volume constraint is formulated as $V(\mathbf{x}) \leq \alpha V_{\max}$. $V_{\max}$ is the maximum volume achieved when all design variables are equal to the upper limit. Boundary conditions (fixed nodes) and external loads are also defined at this stage.
- **Algorithm Parameters:** Algorithm parameters such as the maximum number of iterations (N) and the initial estimate for the Lagrange multiplier (l_{old}) are determined. Additionally, empty data structures such as objective function (‘Cvec’), volume (‘Vvec’), design variables (‘x’ matrix) and Lagrange multiplier (‘l’ vector) are created to store the results at each iteration.

Optimization is performed in a ‘for’ loop until the specified maximum number of iterations is reached or until no significant change in the design occurs. The following steps are performed sequentially at each iteration:

1. **Finite Element Analysis (FEA):** The global stiffness matrix $\mathbf{K}(\mathbf{x})$ of the structure is generated using the design variables ($\mathbf{x}_{old}$) from the current iteration. Then, the nodal displacements ($\mathbf{u}$) are calculated by solving the linear equation system $\mathbf{K}(\mathbf{x})\mathbf{u} = \mathbf{F}$.
2. **Sensitivity Analysis:** The performance of the current design (objective function, $C = \mathbf{F}^T \mathbf{u}$) is calculated. Then, the gradients (sensitivities) of the objective function and volume constraint with respect to the design variables are calculated analytically. The sensitivity of the fitness function with respect to the cross-sectional area (x_j) is found by the formula $\frac{\partial C}{\partial x_j} = -\mathbf{u}_j^T \mathbf{k}_j^0 \mathbf{u}_j$. The sensitivities of the volume constraint are equal to the lengths of the relevant elements (L_j).
3. **Optimization Step (Solution of the Dual Problem):** Using the CONLIN method, a linearized, convex, and separable approximation (sub-problem) of the objective function and constraints around the current design is created. This sub-problem is solved via Lagrange duality. The `fsolve` function in MATLAB is used to find the optimal Lagrange multiplier (λ_{new}) that satisfies the KKT conditions of this dual problem.
4. **Updating Design Variables:** Using the optimal Lagrange multiplier found in the previous step, new values of the design variables ($\mathbf{x}_{new}$) are calculated for the next iteration. These new values are updated to remain within the specified lower and upper bounds (l_b, u_b).
5. **Convergence Check and Saving:** $\mathbf{x}_{new}$ is assigned as $\mathbf{x}_{old}$ for the next iteration. At the end of each iteration, the updated objective function, volume, and Lagrange multiplier values are saved in the corresponding vectors. These data form the basis of the convergence graphs presented in Section 5.

This cycle is repeated until the optimal material distribution is achieved. FEA is the most computationally demanding step in this cycle.

4 Theoretical Methodology

This section details the fundamental theoretical components underlying the optimization process of truss structures. The process is built on three main pillars: (1) Finite Element Analysis (FEA),

which is used to model the structural behavior; (2) the calculation of the gradients, i.e., compliance sensitivities, that drive the optimization algorithm; and (3) the iterative optimization strategy, which combines the CONLIN method with Lagrangian duality.

4.1 Finite Element Analysis of the Truss Structure

Finite Element Analysis (FEA) is a computational technique that models the behavior of a physical object by dividing it into a large number of small, simple elements connected at points called nodes. For truss structures, this method allows for the calculation of nodal displacements under applied loads. A 2D truss structure consists of bar elements connected by hinges, and a structure with N nodes has a total of $2N$ Degrees of Freedom (DOF).

The stiffness of the elements is first defined in their local coordinate systems ($\mathbf{k}_{\text{local}}$), then transformed to a common global reference system via a transformation matrix ($\mathbf{T}_j$) to analyze the structure as a whole:

$$\mathbf{k}_{\text{global}_j} = \mathbf{T}_j^T \mathbf{k}_{\text{local}} \mathbf{T}_j = \frac{Ex_j}{L_j} \begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix} \quad (8)$$

Where E is the Young's Modulus, x_j is the cross-sectional area of the element, L_j is its length, and c and s are the cosine and sine values of the element's angle, respectively. The global stiffness matrix ($\mathbf{K}$) for the entire structure is formed by assembling these element matrices. For a linear elastic problem, the equilibrium equation of the system is as follows:

$$\mathbf{K}(\mathbf{x})\mathbf{u} = \mathbf{F} \quad (9)$$

In this equation, $\mathbf{K}(\mathbf{x})$ represents the global stiffness matrix dependent on the design variables; $\mathbf{u}$ represents the unknown nodal displacement vector; and $\mathbf{F}$ represents the known external load vector.

4.2 Compliance Sensitivity Analysis

The most critical step in gradient-based optimization is the calculation of the derivatives, or sensitivities, of the objective function with respect to the design variables. The compliance (C) of a structure is defined as the work done by external forces: $C(\mathbf{x}) = \mathbf{F}^T \mathbf{u}(\mathbf{x})$. This is equivalent to the total elastic strain energy stored in the structure.

For a truss element with x_j (the cross-sectional area of the j -th element) as the design variable, the compliance sensitivity is given by simplification due to the self-adjoint problem as follows:

$$\frac{\partial C}{\partial x_j} = -\mathbf{u}_j^T \mathbf{k}_j^0 \mathbf{u}_j \quad (10)$$

Here, $\mathbf{u}_j$ is the displacement vector of the j -th element's nodes in the global reference system. $\mathbf{k}_j^0$ is the stiffness matrix for the element with unit cross-sectional area ($\mathbf{k}_{\text{global}_j} = x_j \mathbf{k}_j^0$). The expression on the right-hand side represents the strain energy density of the element and is always positive. Therefore, the sensitivity value is always negative ($\partial C / \partial x_j < 0$). This physically confirms that adding material (increasing the cross-sectional area) will always decrease compliance (increase stiffness).

4.3 Optimization via CONLIN and Lagrangian Duality

Structural optimization problems are often solved by transforming the original complex problem into a series of simpler, more approximate subproblems. Lagrangian duality and KKT conditions provide a fundamental framework for solving these subproblems.

4.3.1 The Primal and Dual Problem Formulation

A general constrained minimization problem (primal problem) is expressed as follows:

$$\begin{aligned} \min_{\mathbf{x}} \quad & g_0(\mathbf{x}) \\ \text{s.t.} \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\ & \mathbf{x} \in X \end{aligned} \quad (11)$$

Here, g_0 represents the objective function (fitness, C in our case), g_i represents the constraint functions (volume constraint in our case), and X represents the box constraints ($l_b \leq x_j \leq u_b$). To solve this problem, the Lagrange function ($\mathcal{L}$) is defined:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = g_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) \quad (12)$$

Here, the vector $\boldsymbol{\lambda}$ contains the non-negative Lagrange multipliers corresponding to each constraint ($\lambda_i \geq 0$). The KKT (Karush-Kuhn-Tucker) conditions define the necessary (and for convex problems, sufficient) conditions for a solution to be optimal. Instead of solving these conditions directly, the dual problem is often solved. The dual function $\phi(\boldsymbol{\lambda})$ is defined as:

$$\phi(\boldsymbol{\lambda}) = \min_{\mathbf{x} \in X} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) \quad (13)$$

The dual problem is the maximization of this function: $\max_{\boldsymbol{\lambda} \geq 0} \phi(\boldsymbol{\lambda})$. For convex problems, the solution of the dual problem gives the solution of the primal problem (strong duality).

4.3.2 The CONLIN Method

Since the original optimization problem is generally non-convex, the CONLIN (Convex Linearization) method transforms the problem into an approximate convex subproblem that is easier to solve with each iteration. CONLIN uses a "hybrid approximation" technique that linearizes a function $g(\mathbf{x})$ around the current design point $\mathbf{x}^k$, either directly in terms of the variable (x_j) or its inverse ($1/x_j$), depending on the sign of its sensitivity. This produces a general form of separable approximation as follows:

$$g^k(\mathbf{x}) \approx \sum_{j=1}^n \left(\frac{a_j^k}{x_j - b_j^k} \right) + c^k \quad (14)$$

Here, the coefficients a_j^k, b_j^k, c^k are derived from the function value and the sensitivities at step k . Since the fitting sensitivity is always negative ($\partial C / \partial x_j < 0$), a more stable approximation to the objective function is often obtained by using inverse variables.

4.3.3 Iterative Solution Procedure

The optimization process consists of a loop in which the following steps are repeated:

1. **Run FEA:** Perform Finite Element Analysis for the current design $\mathbf{x}^k$ to calculate the displacements ($\mathbf{u}$), compliance (C), and volume (V).
2. **Establish CONLIN Approximation:** Calculate the sensitivities of the objective and constraint functions ($\partial C / \partial x_j$ and $\partial V / \partial x_j$) and use these values to construct CONLIN approximations of the functions.
3. **Solve Dual Problem:** For the generated convex and separable subproblem, solve the dual problem to find the optimal Lagrange multiplier (λ^{k+1}) for the next step. This is typically a one-dimensional root-finding problem where $\phi(\boldsymbol{\lambda})$ is maximized and satisfies the KKT conditions.

4. **Update Design Variables:** Using the found value of λ^{k+1} , calculate the new design variables ($\mathbf{x}^{k+1}$) with a closed-form expression derived from the KKT stationarity condition and apply the box constraints. The process is repeated from step 1 until convergence is achieved.

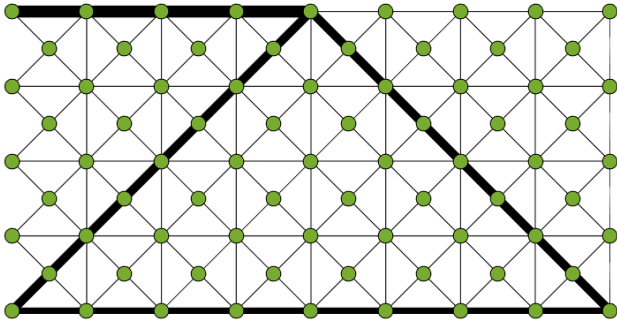
5 Results and Discussion

This section presents and discusses the findings from the optimization process of the 2D truss structure. The optimization was performed for four different volume fraction scenarios ($\alpha = 0.05, 0.10, 0.15, 0.20$), as specified in the project description. The presented results demonstrate the influence of the existing material on the final structural configurations, the convergence behavior of the optimization algorithm, and the stress distribution in critical elements.

5.1 Optimized Topologies for Volume Constraints

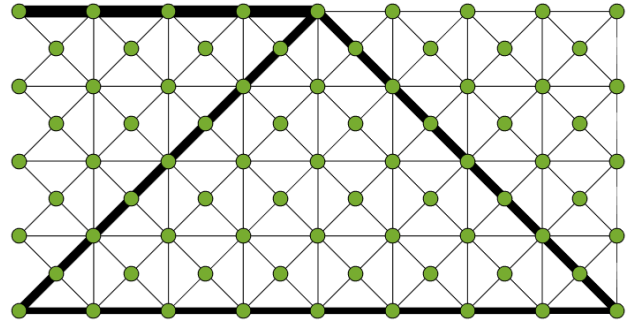
For each volume fraction (α), the CONLIN method and the dual formulation effectively minimize structural compliance while adhering to the global volume and element box constraints. The resulting optimal material distributions reveal a clear trade-off between structural stiffness and material usage.

Optimized Structure $\alpha = 0.05$



(a) Optimized topology for $\alpha = 0.05$.

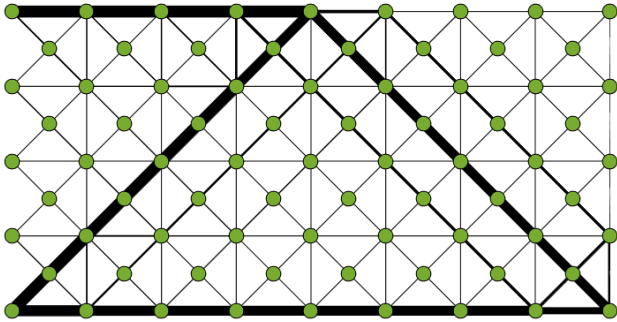
Optimized Structure $\alpha = 0.1$



(b) Optimized topology for $\alpha = 0.10$.

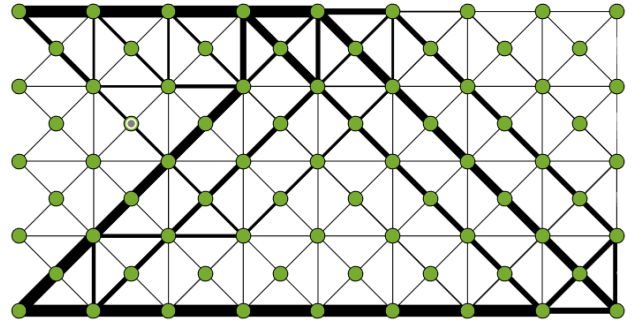
Figure 3: Optimized structural topologies for low volume fractions. Line thicknesses are proportional to the optimized cross-sectional areas.

Optimized Structure $\alpha = 0.15$



(a) Optimized topology for $\alpha = 0.15$.

Optimized Structure $\alpha = 0.2$



(b) Optimized topology for $\alpha = 0.20$.

Figure 4: Optimized structural topologies for high volume fractions.

5.1.1 Discussion of Topologies

When examining the results, the effect of volume fraction on the final topology is clearly evident. At the low volume fraction ($\alpha = 0.05$), the optimization algorithm concentrated material only on the primary load path, which transfers the load to the supports. This resulted in a very sparse structure, with secondary or less critical elements almost completely removed (their areas reduced to a minimum). As the available material increased ($\alpha = 0.10$ and $\alpha = 0.15$), the structure developed stronger load paths and retained or strengthened secondary elements that contributed to stiffness. At the highest volume fraction ($\alpha = 0.20$), the stiffest but also heaviest design was achieved, featuring significantly thicker elements in both the primary and secondary load paths. This demonstrates how increased material utilization can significantly improve structural performance.

5.2 Convergence Analysis

The optimization process exhibited a stable convergence behavior for all tested volume fractions. The evolution of the objective function (fitness) and the material volume over the iterations provides important information about the efficiency of the algorithm and the characteristics of the problem.

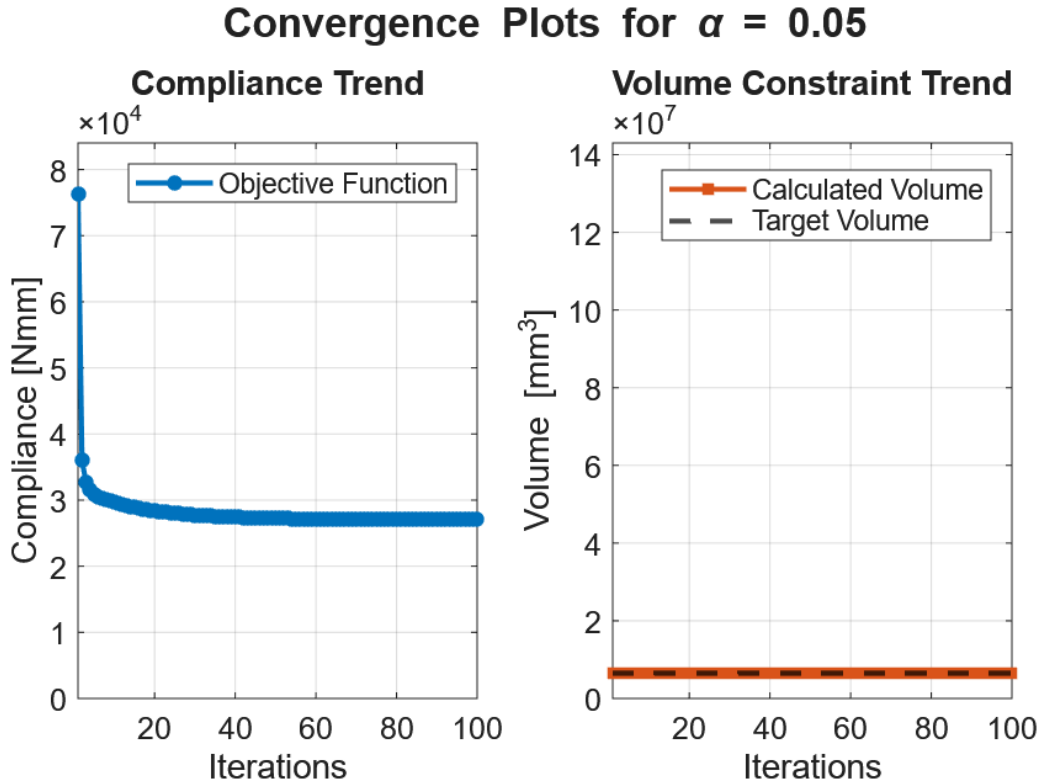


Figure 5: Convergence of the objective function (left) and the volume constraint (right) for the $\alpha = 0.05$ volume fraction scenario

Convergence Plots for $\alpha = 0.1$

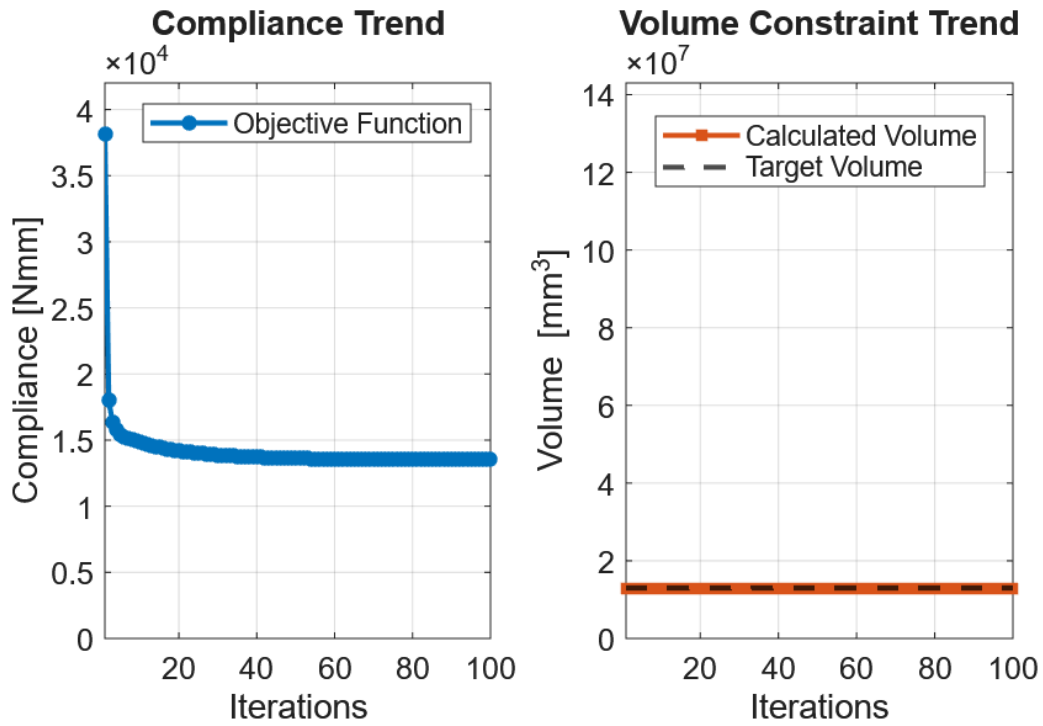


Figure 6: Convergence of the objective function (left) and the volume constraint (right) for the $\alpha = 0.01$ volume fraction scenario

Convergence Plots for $\alpha = 0.15$

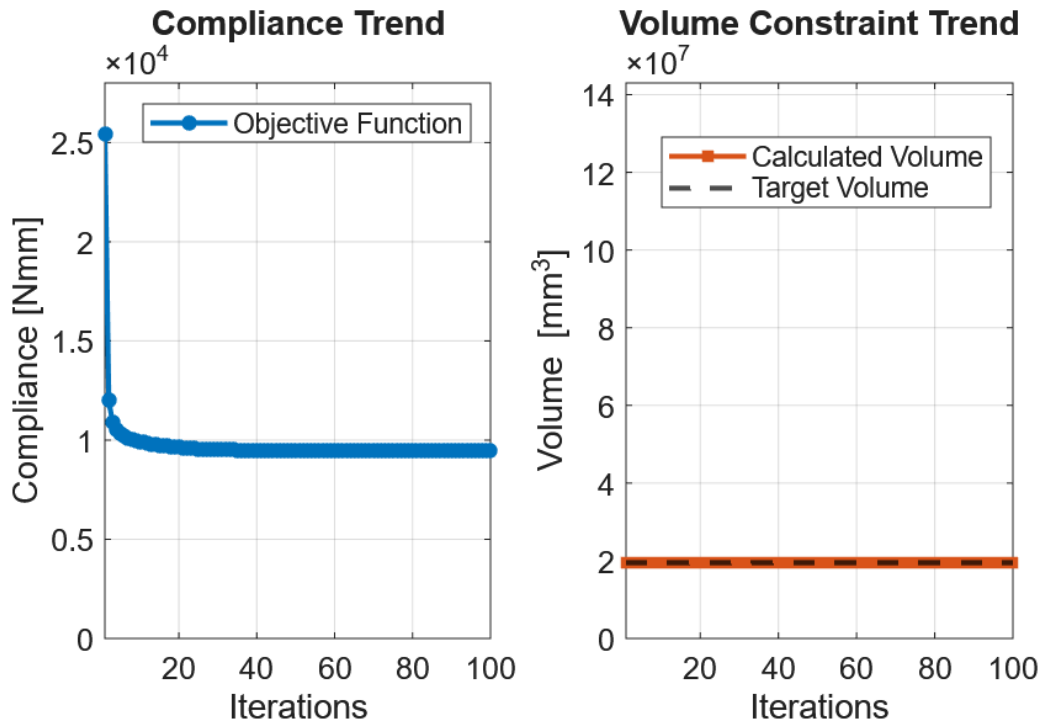


Figure 7: Convergence of the objective function (left) and the volume constraint (right) for the $\alpha = 0.15$ volume fraction scenario

Convergence Plots for $\alpha = 0.2$

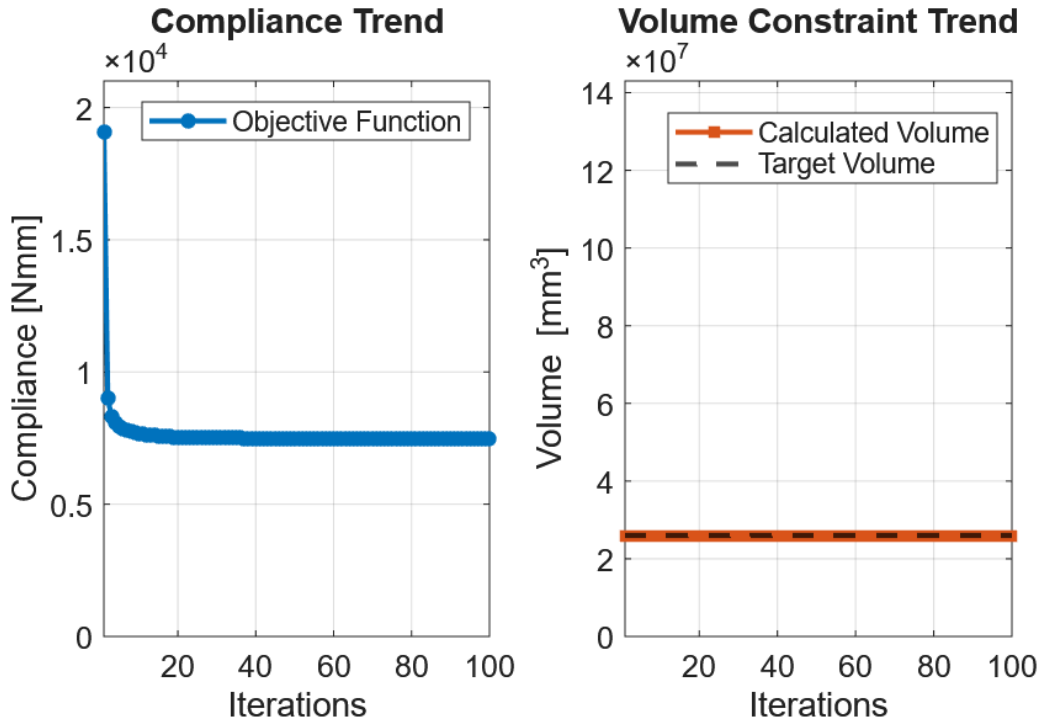


Figure 8: Convergence of the objective function (left) and the volume constraint (right) for the $\alpha = 0.2$ volume fraction scenario

The table below summarizes the final fitness and volume values obtained for each case after 100 iterations.

Table 1: Final values after convergence for different volume fractions.

Volume Ratio (α)	Optimal Compliance (N·mm)	Final Volume (mm ³)
0.05	27018.4	6500390
0.10	13509.9	13000800
0.15	9452.05	19501200
0.20	7483.56	26001500

5.2.1 Discussion of Convergence

An examination of the convergence graphs reveals that compliance values in all scenarios drop rapidly in the first few iterations. This indicates that the algorithm rapidly identifies and strengthens the most critical elements, resulting in significant stiffness improvements. As the optimization progresses, the rate of compliance improvement slows down and converges to a plateau. This is a typical sign that the design is approaching its optimal configuration under the relevant constraints. The volume graph confirms that the algorithm successfully converges the total material volume to the targeted constraint value in all cases.

5.3 Stress Analysis of Selected Elements

To further understand the mechanical behavior of the optimized structures, stress analyses were conducted on elements 64, 77, and 82 specified in the project description. This analysis provides insight into load redistribution and the concept of "fully stressed design."

Table 2: Optimized cross-sectional area and stress values of selected elements (64, 77, 82) for different volume ratios

α	Element 64		Element 77		Element 82	
	Area (mm ²)	Stress (MPa)	Area (mm ²)	Stress (MPa)	Area (mm ²)	Stress (MPa)
0.05	[406.0566]	[29.5444]	[203.0874]	[-29.5439]	[285.8611]	[-29.5514]
0.10	[800.0000]	[14.9948]	[408.0223]	[-14.7050]	[574.3276]	[-14.7088]
0.15	[800.0000]	[14.3802]	[652.7978]	[-9.1912]	[800.0000]	[-9.5036]
0.20	[800.0000]	[13.1461]	[800.0000]	[-7.5000]	[800.0000]	[-7.9792]

5.3.1 Discussion of Stress Results

The results in the table illustrate the effect of the volume constraint on the stress levels in the elements. It is observed that at low volume fractions ($\alpha = 0.05$), critical elements tend to have small cross-sectional areas but operate at very high stress levels. This is consistent with a strategy where minimal material is used to carry the load and these elements are stressed close to their material limits. As the volume fraction increases, the cross-sectional areas of these critical elements generally increase, and the stress levels correspondingly decrease. It appears that when more material is available, the optimizer attempts to achieve a lower and safer stress state by distributing the load more uniformly. This behavior is an approximation of the concept of a "fully stressed design," in which most active elements operate at a given stress level.

6 Conclusion

This study investigated the applicability of topology optimization to find the appropriate material distribution to increase the stiffness of a 2D truss structure. The cross-sectional areas of the truss elements were used as design variables, and the objective was to minimize structural compliance. The optimization process was subject to a global volume constraint and separate upper/lower bounds for each design variable. The chosen solution approach was the Convex Linearization (CONLIN) method, which generates convex subproblems solved by dual formulations. Four different material availability scenarios were evaluated within the scope of the study: 5The key results obtained can be summarized as follows:

- **Optimized Topologies and Volume Constraints:** Optimized structures obtained for four different volume ratios (α) clearly demonstrated the direct impact of the amount of material present on the final geometry and structural performance. At lower α values (e.g., 5
- **Convergence Analysis:** The optimization process based on the CONLIN method demonstrated robust and smooth convergence across all volume fraction scenarios. The compliance values initially decreased rapidly, then the rate of decrease slowed as the optimal solution was approached. It was observed that the volume constraint was consistently met or saturated in the final design, an expected result of the dual formulation.
- **Stress Analysis of Selected Elements:** By examining the cross-sectional areas and corresponding stress levels of specific truss elements (e.g., Elements 64, 77, and 82), the load distribution and the concept of "fully stressed design" were better understood. At low values of α , critical elements generally had smaller cross-sectional areas but operated at high stress levels. As α increased, the areas of these elements generally increased and their stresses decreased, resulting in a more uniform stress distribution. This demonstrates that the optimization balances material efficiency with structural integrity.

In conclusion, the CONLIN method proved to be an effective tool for minimizing the compliance of 2D truss structures under global volume constraints and element size constraints. The combination of Lagrangian duality and separable convexity approaches provided a robust and computationally efficient solution process that converged smoothly in all tested scenarios. Increasing the allowable volume fraction consistently produced stiffer structures with lower compliance and additional supports, clearly demonstrating the trade-off between material usage and structural performance.